Metrics for Evaluating Performance

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Based on slides created by Mark Hill and others

Performance

- Which computer is fastest?
- Not so simple

Example of airplane

Airplane	Passenger capacity	Cruising range (miles)	Cruising speed (m.p.h.)	Passenger throughput (passengers × m.p.h.)
Boeing 777	375	4630	610	228,750
Boeing 747	470	4150	610	286,700
BAC/Sud Concorde	132	4000	1350	178,200
Douglas DC-8-50	146	8720	544	79,424

FIGURE 1.14 The capacity, range, and speed for a number of commercial airplanes.
The last column shows the rate at which the airplane transports passengers, which is the capacity times the cruising speed (ignoring range and takeoff and landing times).

- Plane with the highest cruising speed was the Concorde, the plane with the longest range is the DC-8, and the plane with the largest capacity is the 747

Throughput and Latency: Definitions

- Throughput
  - Total work done in a given time
- Latency or response time
  - Time between start and completion of an event

Latency, bandwidth and throughput: Analogy

- **Latency** is the amount of time it takes to travel through the tube.
- **Bandwidth** is how wide the tube is.
- The rate of water flow is the **Throughput**

https://stackoverflow.com/questions/36949735/what-is-the-difference-between-latency-bandwidth-and-throughput

Solved Example

- ProcessorA: Has 4 cores. Every task takes 5 seconds.
- ProcessorB: Has 2 cores. Every task takes 4 seconds.

- Q1. Which processor has lower latency?
- Answer: ProcessorB.
- Q2. In 100 seconds, which processor will complete more tasks?
- Answer:
- ProcessorA will finish 20*4=80 tasks
- ProcessorB will finish 25*2=50 tasks.
- ProcessorA has higher throughput.
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Latency vs. Throughput

- Is throughput = 1/latency?
  - Only if NO overlap
  - Otherwise, throughput > 1/latency

Lets understand this. Throughput is average number of units processed per time unit. Lets say, units here are instructions.

So, throughput is the number of instructions completed per second.

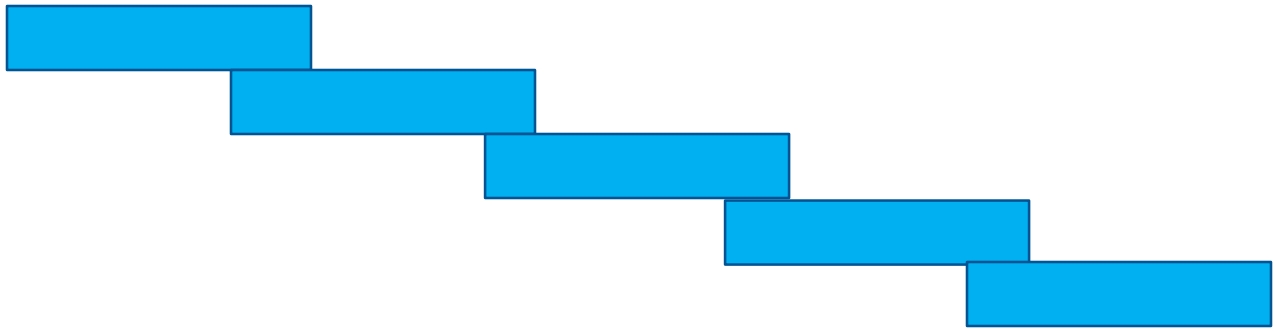
Let the latency of each instruction be 5 milli-second.

Case 1

- **Assume there is no overlap between the instruction-execution**, so different instructions execute serially.
- So, in one second, we will execute 1/(5*10e-3) = 200 instructions.
- So throughput is 200 instructions/second. And latency is 5 milli-second.
- So, they are inversely related.

Case 2

- There is overlap between instruction-execution.
- Assume an instruction starts when the previous instruction is 80% done.



- Clearly, if number of instructions executed in 1 second is (x+1), then we have 5+4x=1000.
- Reason: first instruction will take 5 ms effective latency and all remaining instructions will have an effective latency of 4 ms each.

Case 2 (continued)

- Solving, we get x=248. So, throughput is 249. So, although inverse of latency is only 200, but the throughput is 249, so throughput is larger than inverse of latency.
- If we assume that an instruction starts when the previous instruction is 50% done, then 5+2.5x=1000 so, x=398. So throughput becomes 399 instructions/second. So, since the overlap has increased, the throughput has also increased.

Relative Performance

Example

If computer A runs a program in 10 seconds and computer B runs the same program in 15 seconds, how much faster is A than B?

Answer

We know that A is n times as fast as B if

$$\frac{\text{Performance}_A}{\text{Performance}_B} = \frac{\text{Execution time}_B}{\text{Execution time}_A} = n$$

Thus the performance ratio is

$$\frac{15}{10} = 1.5$$

and A is therefore 1.5 times as fast as B.

Solved question

We have a road from city A to B, with a width of 100 feet. A car moves at a speed of 1000 feet/second. To move safely, a car requires 10 feet space (width). On a weekend, we have only 5 cars running at any point of time, in parallel. Total distance between A and B is 100,000 feet.

Now, show the latency, bandwidth and throughput.

Latency = 100,000/1000 = 100 seconds

Bandwidth = 10 lanes

Throughput = 5 cars

CPU Performance and Its Factors

CPU Performance and Its Factors

$$\text{CPU time} = \text{CPU clock cycles for a program} \times \text{Clock cycle time}$$

$$\text{CPU time} = \frac{\text{CPU clock cycles for a program}}{\text{Clock rate}}$$

$$\text{CPI} = \frac{\text{CPU clock cycles for a program}}{\text{Instruction count}}$$

$$\text{CPU time} = \text{Instruction count} \times \text{Cycles per instruction} \times \text{Clock cycle time}$$

$$\frac{\text{Instructions}}{\text{Program}} \times \frac{\text{Clock cycles}}{\text{Instruction}} \times \frac{\text{Seconds}}{\text{Clock cycle}} = \frac{\text{Seconds}}{\text{Program}} = \text{CPU time}$$

Components of performance	Units of measure
CPU execution time for a program	Seconds for the program
Instruction count	Instructions executed for the program
Clock cycles per instruction (CPI)	Average number of clock cycles per instruction
Clock cycle time	Seconds per clock cycle

Finding time from average CPI

- Different instruction types having different CPIs

$$\text{CPU clock cycles} = \sum_{i=1}^{n} \text{IC}_i \times \text{CPI}_i$$

$$\text{CPU time} = \left(\sum_{i=1}^{n} \text{IC}_i \times \text{CPI}_i \right) \times \text{Clock cycle time}$$

IC = instruction count

Question 1.6

- Consider two different implementations of the same instruction set architecture. The instructions can be divided into four classes according to their CPI (classes A, B, C, and D).
- P1 with a clock rate of 2.5 GHz and CPIs of 1, 2, 3, and 3,
- P2 with a clock rate of 3 GHz and CPIs of 2, 2, 2, and 2.

- Given a program with a dynamic instruction count of 1.0E6 instructions divided into classes as follows: 10% class A, 20% class B, 50% class C, and 20% class D, which is faster: P1 or P2?
- a. What is the global CPI for each implementation?
- b. Find the clock cycles required in both cases.

Solution

- Class A: 10^5 instructions, and so on.
- Time = (NumInstruction* CPI)/ClockRate
- TotalTime_P1 =
- (10^5 + 2*10^5*2 + 5*10^5*3 + 2*10^5*3)/(2.5*10^9)
- = 10.4* 10^-4 seconds
- TotalTime_P2 = 6.66 * 10^-4 seconds

- (a) Global CPI is weighted average of CPI values.
- CPI_P1 = Time*ClockRate/NumInstruction = 2.6
- CPI_P2 = 2.0

Solution

$$\text{CPU clock cycles} = \sum_{i=1}^{n} \text{IC}_i \times \text{CPI}_i$$

- ClockCycles_P1
- = 10^5 + 2*10^5*2 + 5*10^5*3 + 2*10^5*3 = 26 * 10^5

- ClockCycles_P2
- = 20 * 10^5

Measure of Performance

$$\frac{\text{Time}}{\text{Program}} = \frac{\text{Instructions}}{\text{Program}} \times \frac{\text{Cycles}}{\text{Instruction}} \times \frac{\text{Time}}{\text{Cycle}}$$

(code size) (CPI) (cycle time)

Architecture --> Implementation --> Realization

Compiler Designer **Processor Designer** **Chip Designer**

Note: Performance is inverse of time. More the execution time, less the performance

Our Goal

- Minimize time which is the product, NOT isolated terms
- Common error to miss terms while devising optimizations
  - E.g. Suppose we change the ISA, such that the instruction count is decreased
  - BUT it may lead to a CPU organization which makes the clock slower
- Bottom line: terms are inter-related

ISA = instruction set architecture

- Is a 4 GHz machine faster than a 2 GHz machine? Justify your answer.
- Answer: Not necessarily.
- Reason: performance of processor does not depend on clock frequency alone. It also depends on the IPC, and the number of instructions. The IPC depends on the compiler, and the architecture which includes the number of pipeline stages.

Other Metrics and their limitations

MIPS

- MIPS = instruction count/(execution time x 10^6)

- CPI = ClockCycles/InstructionCount
- ➔ InstructionCount = ClockCycles/CPI

- ExecutionTime = ClockCycles/ClockRate

- ➔ MIPS = ClockRate/(CPI x 10^6)

- But MIPS has serious shortcomings

How MIPS can be a misleading metric

- Example: An application computes "sine" function 10 times. On a CPU, a single execution of sine function requires 50 instructions. Overall execution time of the program is 5 micro-seconds.
- (a) Find MIPS of this program.
- Answer: (10*50)/(5*10e-6* 10e6) = 100
- (b) Now, on this CPU, we implement a dedicated engine for computing sine function. This engine computes sine function with a single instruction. Overall execution time of this program is now 2us. Find MIPS of this program on this upgraded CPU.
- Answer: (10*1)/(2*10e-6* 10e6) = 5
- Comments: On adding the specialized hardware for sine function, the total execution time has reduced, which means performance has improved.
- However, MIPS has reduced. So, if we were to use MIPS as the performance metric, we would think that performance has reduced, which is misleading.

Problems with MIPS (and MFLOPS)

- It ignores program

- When is MIPS ok?
  - Same compiler, same ISA
  - E.g. same binary running on Pentium-III, IV
  - Why? Instr/program is constant and can be ignored

- MFLOPS = FP operations in program/(execution time x 10^6)
- MFLOPS can be used if FP ops remain same.

Op= operation, FP = floating-point